

2. Uniform Source

This assumes the whole area of the star emits lights equally. So essentially

$$F = 1 - \frac{\text{Area Overlap}}{\pi r_*^2}$$

In our model everything is normalised so only the relative sizes of two bodies matter (p) and their relative distance between each other matters (z).

Therefore we can write $F^e(p, z) = 1 - I^e(p, z)$, where e stands for eclipse, and I^e is overlap function.

① Firstly, there is no overlap if

$$d > r_p + r_* \quad \text{or} \quad z r_* > p r_* + r_*, \\ \text{ie} \quad p + 1 < z$$

So $F = 1$ here.

② Another easy case is when the planet is fully inside the star.

$$d \leq r_* - r_p \quad \text{or} \quad \frac{z}{r_*} \leq r_* - p r_*, \\ \text{ie} \quad z \leq 1 - p \quad \text{with } p < 1$$

$$\frac{\text{Area Overlap}}{\pi r_*^2} = \frac{\pi r_p^2}{\pi r_*^2} = p^2$$

So $\lambda^e(z, p) = p^2$ and $F^e(z, p) = 1 - p^2$

③ Last easy case is when the star is fully inside the star.

$$d \leq r_p - r_*$$

or $z \leq p - 1$ with $p \geq 1$

$$\frac{\text{Area Overlap}}{\pi r_*^2} = \frac{\pi r_*^2}{\pi r_*^2} = 1$$

So $\lambda^e(z, p) = 1$ and $F^e(z, p) = 0$

④ Most thorough case is when they partially overlap, and whether $p > 1$ or ≤ 1 doesn't matter. It's the only remaining case so:

$$z > 1 - p > 0 \text{ and } z > p - 1 \geq 0 \text{ and } z \leq 1 + p$$

$p < 1$ $p \geq 1$

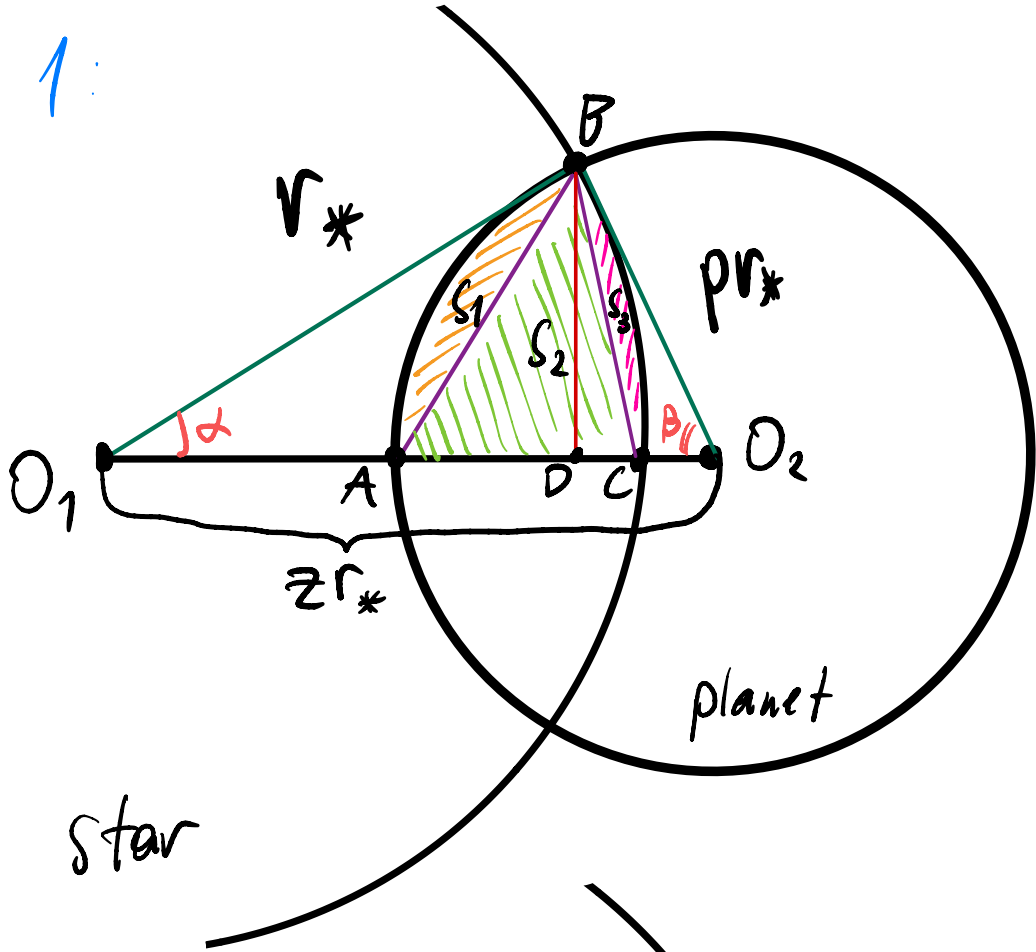


$$z > |1 - p| \text{ and } z \leq 1 + p$$

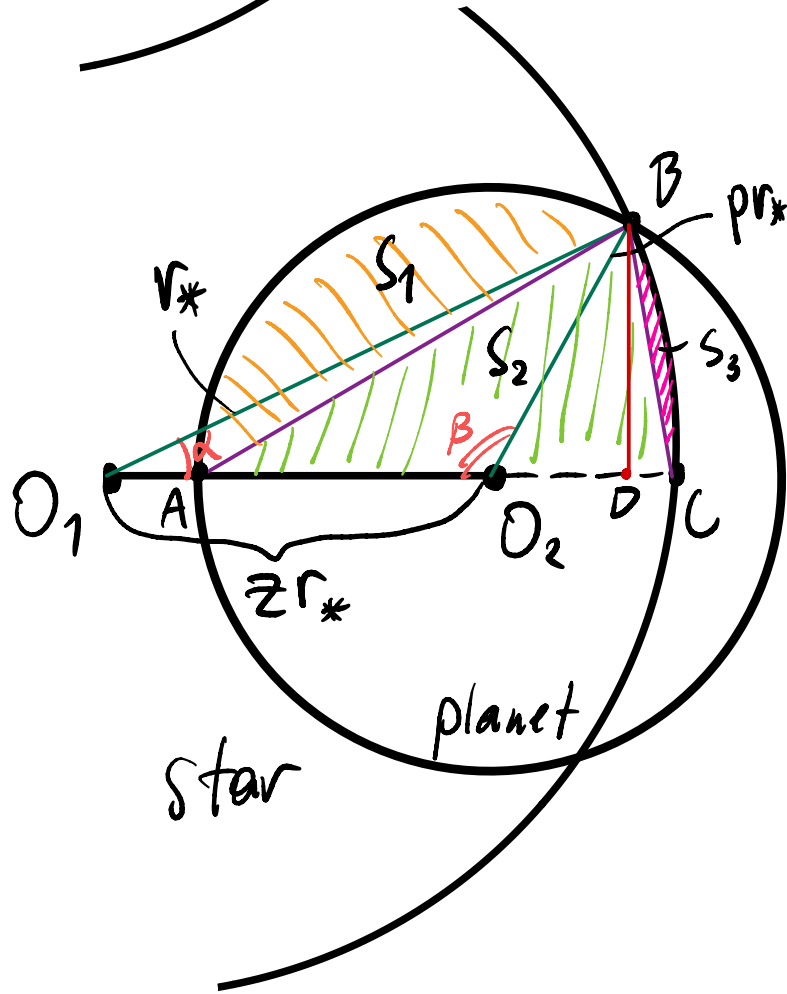
$$\Rightarrow |1 - p| < z \leq 1 + p$$

There are 3 cases with a shared configuration $S_1 + S_2 + S_3$:

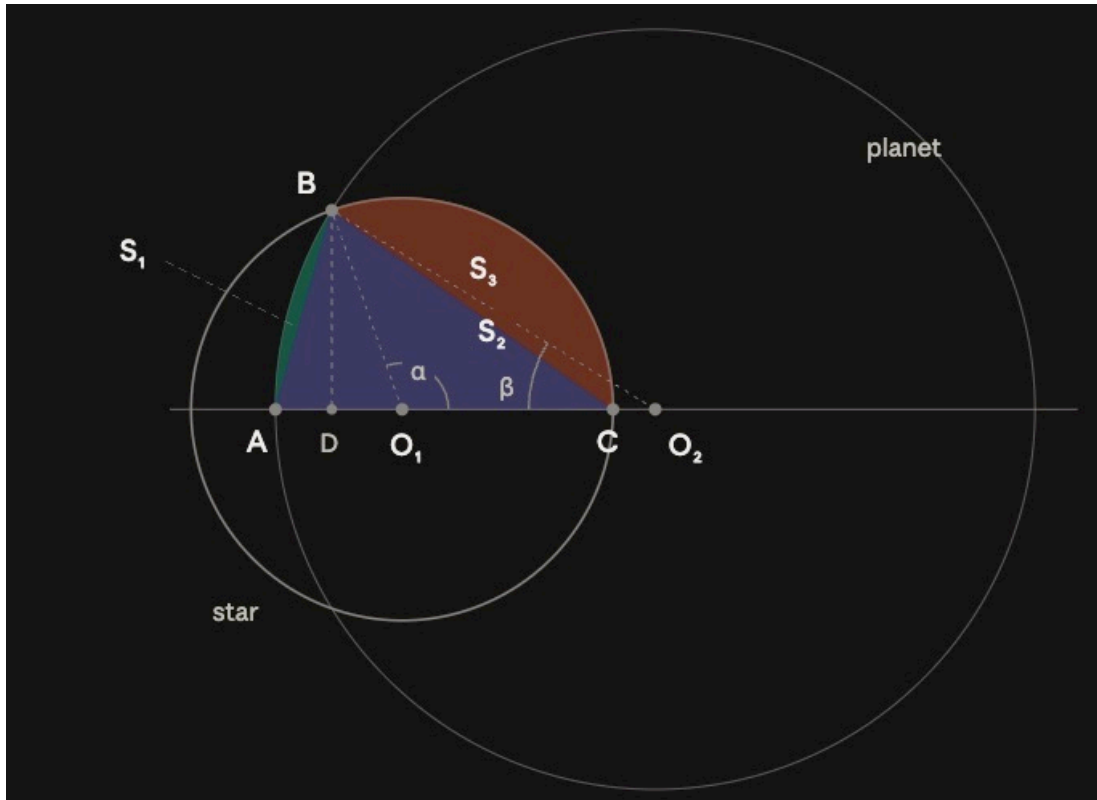
Case 1:



Case 2:



Case 3 (when $p > 1$)



Cosine rule :

$$1) \quad (pr_*)^2 = r_*^2 + z^2 r_*^2 - 2zr_*^2 \cdot \cos \alpha$$

$$\cos \alpha = \frac{z^2 - p^2 + 1}{2z}$$

$$2) \quad r_*^2 = p^2 r_*^2 + z^2 r_*^2 - 2pz r_*^2 \cos \beta$$

$$\cos \beta = \frac{z^2 + p^2 - 1}{2pz}$$

$$\Rightarrow \begin{cases} \alpha = \cos^{-1} \left(\frac{z^2 - p^2 + 1}{2z} \right) \\ \beta = \cos^{-1} \left(\frac{z^2 + p^2 - 1}{2pz} \right) \end{cases}$$

$$BD = r_* \sin \alpha$$

$$S_2 = \frac{1}{2} BD \cdot AC$$

$$= \frac{1}{2} r_* \sin \alpha (2r_* - (2r_* - pr_*) - (2r_* - r_*))$$

$$= \frac{1}{2} r_* \sin \alpha (r_* + pr_* - 2r_*)$$

$$S_1 = \pi p^2 r_*^2 \frac{\beta}{2\pi} - \frac{1}{2} p^2 r_*^2 \sin \beta$$

$$= \frac{1}{2} p^2 r_*^2 (\beta - \sin \beta)$$

$$S_3 = \frac{1}{2} r_*^2 (\alpha - \sin \alpha)$$

Hence

$$I^2(z, p) = \frac{2(S_1 + S_2 + S_3)}{\pi r_*^2} = \frac{2}{\pi} \left(\frac{1}{2} p^2 (\beta - \sin \beta) \right.$$

$$\left. + \frac{1}{2} (\alpha - \sin \alpha) + \frac{1}{2} \sin \alpha (p - z + 1) \right) =$$

$$\left[\begin{array}{l} \sin \alpha = \sqrt{1 - \cos^2 \alpha} \\ \sin \beta = \frac{\sin \alpha}{p} \end{array} \right]$$

$$= \frac{2}{\pi} \left(\frac{1}{2} p^2 \beta + \frac{1}{2} \alpha - \frac{1}{2} p \sin \alpha + \frac{1}{2} \sin \alpha (p - z) \right)$$

$$= \frac{2}{\pi} \left(\frac{1}{2} p^2 \beta + \frac{1}{2} \alpha - \frac{1}{2} \sqrt{\frac{4z^2 - (z^2 - p^2 + 1)}{4}} \right)$$

$$= \frac{1}{\pi} \left(p^2 \beta + \alpha - \sqrt{\frac{4z^2 - (z^2 - p^2 + 1)}{4}} \right)$$

Hence we get the desired result

$$F^c(p, z) = 1 - \lambda^c(p, z) \quad \text{where}$$

$$\lambda^c(p, z) = \begin{cases} 0 & 1+p < z \\ p^2 & 1-p \geq z \\ 1 & p-1 \geq z \\ \frac{1}{\pi} \left[p^2 \beta + \alpha - \sqrt{\frac{4z^2 - (z^2 - p^2 + 1)}{4}} \right] & |p-1| \geq z > 1+p \end{cases}$$

$$\text{and } \alpha = \cos^{-1}\left(\frac{z^2 - p^2 + 1}{2z}\right), \quad \beta = \cos^{-1}\left(\frac{z^2 + p^2 - 1}{2pz}\right)$$